TKELL KNIVES · DESIGN STUDIO

Installation and user guide

Version 1.3.0 · For Windows and Mac · Send this with the program

> **New in 1.3.0.** Circle is a center and a radius. Type that radius, or drag the arc. Any other curve can be typed the same way. Tangent makes a curve leave along the line it meets. A plain click on an endpoint draws the next line. Hold Alt (Option on a Mac) to drag the point. Hold Alt on a gold dot to move that end of the tang.

1. Getting the program

The package is the drawing studio plus the handle and blade master files. It opens in Chrome. It is not a file you double-click.

Download the newest package here. This page is public, so the link can be mailed. It does not ask for a GitHub login:

https://github.com/JPFava/tkell-studio-share/releases/latest

Take both files. This PDF opens by itself. The program is the zip. The zip asks for a password. That password is in the note from TKell. It is not written on the download page.

> **Do not attach the zip to Gmail.** Gmail marks the program “Dangerous” and refuses it, because the folder contains the studio’s code. Send the link above and the password as ordinary text. The person downloads the files themselves.

> **The STEP kernel is not inside the zip.** Drawing, scans, save, and load work as soon as the studio is running. Build, STEP, and DXF also need Build123d, the OpenCASCADE modeler. Install it once on that same Windows or Mac computer. The steps are in Appendix A.

2. Install

What to install first

1. Google Chrome.
2. Node.js 22, from nodejs.org. Take the current 22 release. During the Windows install, leave “Add to PATH” checked.

Windows

1. Unzip `TKell-Design-Studio.zip`. Put the folder somewhere you can keep, such as Documents. Do not run it from inside the zip.
2. Open the folder. Click the address bar, type `cmd`, and press Enter. A command window opens in that folder.

3. Type `npm install` and press Enter. Wait until it finishes. This only has to be done once, and again after a new version.
4. Type `npm run dev` and press Enter. Leave that window open.
5. Open Chrome and go to `http://127.0.0.1:8080`

Mac

1. Unzip the package and move the folder out of Downloads if you want to keep it.
2. Open Terminal. Type `cd ` and then drag the folder onto the Terminal window. Press Enter.
3. Type `npm install` and press Enter.
4. Type `npm run dev` and press Enter. Leave Terminal open.
5. Open Chrome and go to `http://127.0.0.1:8080`

> To stop the studio, click the command window and press Ctrl+C (Windows) or Control+C (Mac). Next time you only need `npm run dev`, then Chrome. You do not run `npm install` again unless you replaced the folder with a newer package.

3. The window

The tools sit in one bar across the top. Under that is a single view. **2D** is the drawing. **3D** is the solid. The toggle is at the right end of the bar. Only one of them is on screen.

File, Handle, Scan, Chat, and Features are closed until you click them. They close when you pick an item, click Close, or click the drawing. A feature you are editing stays open until you close it.

ON THE BAR	WHAT IT DOES
Line, Circle, Select, Trim, Extend, Delete, Fillet, Tangent, Break, Move	Draw and edit the outline. Circle is a true radius. Fillet rounds a corner. Tangent makes a curve leave along the line it meets. The Fillet field is on the bar only while Fillet is the tool. The Radius field appears when a curve or a circle is selected.
Choil, Ramp, Jimping, Bevel, Swedge, Chamfer, Edge fillet, Fuller	Knife features. Click the tool, then the line it belongs on. Edge fillet is the round version of the chamfer.
Snap	On sticks to endpoints, intersections, and construction points. Off lets you place freely.
Undo, Redo	One action at a time. This is not the same as Restore.
Fit	Zooms the 2D drawing to the knife.
File	Blade name, presets, save, export, import, clear lines, restore.
Handle	The grip the blade is drawn on.
Scan	A photo or drawing under the sketch.
Chat	Type a change in words.
Features	Open a feature you already placed, so you can edit it.
Stock	Blade thickness in inches. Default is 0.165.
Build	Writes the STEP and shows it in 3D. Stays dim until the outline is one closed loop.
STEP, DXF	Download the last blank that Build wrote.

The line under the bar is the status. Read it. It tells you what the next click is supposed to do.

4. Start a knife

1. Click **Handle** and pick the grip. The handle and its tang land on the 2D board, tip direction to the right. The view zooms out past the handle so there is room for a blade.
2. Click **File**. Set the blade name. That name is used on the saved JSON, the STEP, and the DXF.
3. Still in File, you can start from a preset instead of a blank tang: Drop point, Dagger, Wharncliffe, Tanto, Clip point, or Sheepsfoot. The preset is a sketch on the current handle. You can still edit every line.

The handle you pick is the master grip. Changing handles does not throw away lines you already drew. If the new tang does not meet those lines, use **Move** (below) to shift the drawn blade onto the new tang.

5. Draw the outline

Stay on **2D**. Units on the board are inches.

Line

Click two points. That is a line. Click the next point to continue. A click on an endpoint starts the next line from that point. To move the endpoint instead, hold Alt (Option on a Mac), click the point, and keep holding Alt while you drag. The cursor shows which one you are about to do. The draw cursor starts a line. The move cursor drags the point.

Grab the middle of a line and drag to bow it into a curve. The Radius field appears on the bar. Type the radius in inches, or keep dragging. The curve stays on the side you dragged. A radius shorter than half the distance between the two ends is refused. Those two ends cannot make a curve that tight.

The two gold dots are the ends of the tang. Hold Alt and drag a gold dot to move that end. A sketch point that is already sitting on that end moves with it.

Circle

Click the center, then drag out the radius, or click the center and click again where the rim should be. The Radius field is on the bar. Type the radius, or drag any arc later to change it. Hold Alt and drag a point on the circle to move the whole circle. This is a true radius. Do not try to fake one by bowing two lines against each other.

Select

Click a line to select it. Drag a box to take every line whose end or middle falls inside the box. Hold Shift while you click or drag to add to the lines already selected. A click on empty paper clears the selection. Selected lines are blue. Press Delete or Backspace to remove them. When one curve or one circle is selected, its radius is on the bar and you can type a new one. With Select on, Shift does not pan the view. Use the middle mouse button, or the scroll wheel, to move around.

Trim

Click the piece of a line you want gone. The piece is the stretch between crossings with other lines. A click near an open end shortens that end back to the crossing. A click between two crossings takes out the middle and leaves the two sides. If the line does not cross another line, there is nothing to trim to.

Extend

Click an open end. The line runs straight, on its own direction, until it hits another line. An end that already meets another line will not extend. Use this to close a gap. The status says when the end has nowhere to land.

Delete

Click a line to delete that line. If that line is selected, the whole selection goes. A connected run of two or more lines becomes one straight line between the two ends that remain. One line just opens. Deleting every line of a closed loop does not draw a new line across the gap.

Fillet

This rounds a corner of the drawing. It is not a grind, and it does change the outline. Click the corner where two lines meet, or select those two lines and then click. The radius is the Fillet field on the bar. It shows only while this tool is on. The default is 0.060 in. The new piece is an arc. Drag the middle of that arc later, or select it and type a radius, the same way you edit any other curve. If the radius is longer than either line, type a smaller one. A corner where more than two lines meet needs those two lines selected first.

Tangent

Use this when a curve should leave along the line or curve it already meets. Click near that end. The curve adjusts, and the new radius shows in the Radius field. A corner that is too square cannot be made tangent with one curve. The status says so. Use Fillet for that corner. If a circle does not already meet the line, click the circle, then click the line. The circle slides until it sits tangent, and it keeps its radius.

Break

Click a line or curve where you want a joint. One line becomes two, and they share that point, so the outline stays connected.

Move

Drag a box around the points you want, then pick a start point and an end point. Those points can be snaps. With no box, the next two clicks move everything you drew, from the first snap to the second. Use this after a handle change that leaves a gap at the tang.

Snap

Leave Snap on while you close the outline and while you land a feature on a corner. Turn it off when a click keeps jumping to the wrong point.

When Build will light up

Build stays unavailable while the outline is open. The button shows how many open loops are left. You need one closed outline. A closed loop drawn inside that outline is a cutout. An open loop that starts and ends on the outline and bulges outside it is an addition, such as a ring or a glass breaker. The glass breaker on the Agent tang is part of the handle master; do not trim it off when you only meant to change the blade.

Undo and **Redo** step one action. **Restore**, under File, puts the knife back to the starting sketch. It is not an undo.

6. Features

Features stay attached to the line you picked, so you can open them later from **Features** and change the numbers. Choil and thumb ramp are cut out of the blade. They are not material added on the outside.

TOOL	HOW YOU PLACE IT	WHAT YOU SET
Choil	One click on a line.	Radius. An X marker stays on the line and can be dragged.
Ramp	One click on a line.	Radius and height of the thumb ramp.
Jimping	Two clicks, the start and the end of the run.	Radius, how far the grooves sit below the line, and the count. Each groove has sides tangent to the radius, so a flat shows between scallops. Two X markers mark the ends.
Bevel	Two clicks along the edge. The run can cross several connected segments. The tip is usually inside the bevel.	See the bevel section below.
Swedge	Two clicks along the spine.	Height at one end, zero at the other. The tall end can be the start or the stop. If a bevel is in the way, the swedge stops where they meet. This is a triangle, not a flat grind.
Chamfer	Two clicks along a corner.	Angle, default 45°, plus a height at the start and a height at the stop. 45° takes the same amount off the face and into the thickness. This flattens the corner. It does not change the outline.
Edge fillet	Two clicks, the start and the stop, the same way as a chamfer. The run can cross several connected segments.	Radius at the start and radius at the stop. Default is 0.030 in at both ends. This is a quarter-round: the radius is on the face and the same distance into the thickness. The two ends can differ. It does not change the outline. Open the feature to set the radii. Build to see the round in the solid.
Fuller	Draw a closed loop inside the outline, then click Fuller and that loop.	Depth and the radius in the bottom. Depth 0 makes it a through cutout again.

Bevel

Defaults are a 20° primary, a 40° secondary, an edge flat of 0.018 in, and a secondary meet of 0.040 in. Both angles are **inclusive**. A 20° primary is 10° each side of center, not 20° each side.

- **Edge flat** is the thickness left at the cutting edge. The primary starts on center, set back by half of that flat, on both sides.
- **Sharpen** is on by default. Turn it off for a flat grind with no secondary bevel. A swedge is a separate tool. Do not fake a swedge by turning Sharpen off.
- **Secondary by angle** runs the secondary out from center until it meets the primary. The panel tells you the height where they meet.
- **Secondary by height** makes them meet at the distance you type, and the panel tells you the angle that takes.

- The bevel height stays parallel to the edge. It does not shrink just because the tip is near. If the blade runs out, the grind runs out with it.
- **Plunge radius** is a ball-end mill held square to the face. The radius is constant. The depth goes from full at the bevel to zero where the grind meets the flat. Plunge angle is from square: −5° to +10°. Zero is perpendicular. Positive leans toward the tip.
- Where two bevels continue into each other, there is no plunge between them. The plunge is only at the end that is not the tip.

The 2D view shows the bevel landing at full thickness. The 3D view, after Build, is the grind in the STEP, not a picture painted on the blank.

Chamfer and edge fillet

Both sit on a corner of the outline. Neither one moves the line you drew. A chamfer takes a flat off that corner. An edge fillet puts a quarter-round there instead. Set the angle and the two heights on a chamfer. Set the two radii on an edge fillet. Neither one has a plunge. In the drawing, a chamfer is a green dashed mark and an edge fillet is a lilac dashed mark. The solid shows them only after Build.

7. Build, STEP, and DXF

1. Close the outline. Build is clickable when it no longer shows an open-loop count.
2. Set **Stock** if this blade is not 0.165 in.
3. Click **Build**. The view switches to 3D. The status counts the seconds. Leave it alone until it finishes. Clicking Build again starts the job over.
4. The 3D view is orthographic. Drag to orbit. The lamp can be dragged so a swedge or a fuller is easier to see. Click **2D** to go back to the drawing. The solid stays.
5. **STEP** downloads the blade blank. **DXF** downloads the laser profile. Both use the blade name.

> If Build says Build123d is not installed, finish Appendix A, then click Build again.

8. A scan or a sketch under the knife

Click **Scan**, then **Load scan**. Photos, PNG, JPG, WEBP, and iPhone HEIC files are accepted.

1. Click **Match**. On the picture, click a point or drag a box around a hole. Do the same for a second hole. Then pick the same two holes on the handle model, in the same order. With Snap on, the center of a handle hole will catch the click.
2. Which end you pick first does not have to be a particular hole, but the second pair must be the same two holes in the same order.
3. If the artwork is left-handed and the studio handle is the usual right-handed view (handle left, tip right), turn **Mirror** on before you match, or the drawing will land upside down.
4. A box around a hole looks for one circle. If it fails, the status says why: not enough contrast, not round enough, or more than one circle. A few stray line bits are tolerated. A second full circle in the box is not.

Hand drawings will not scale perfectly. After the match, nudge about the first handle point. Each control is independent, and you can go back and forth:

- Rotate in steps of 0.01°, limited to ±2.00°.
- Shift one picture pixel at a time. Hold the button to keep moving.
- Scale from 0.900× to 1.100× in steps of 0.001.

Undo nudge and Redo nudge only undo those nudges. They do not undo the knife. Holes help, but the place the drawn blade meets the handle matters more. Nudge until that junction is right.

Construction geometry

After the scan is matched and Show scan is on, click **Add construction geometry** and drag a box around ink you want to trace. Lines, curves, and endpoints in that box become snap points. They are not part of the knife. Show construction turns them on and off. They move with later nudges. Use them to draw a cutout or an extension that follows the artwork. This command is only there after a scan is placed.

9. Save, export, and send a design to someone else

Under **File**:

- **Save** keeps the blade name, the handle, and the sketch on this computer.
- **Export** writes a JSON file. It includes the sketch, the scan picture, the hole match, the nudges, and the construction geometry. The file name includes the blade name. This is the file to email another designer.
- **Import** opens that JSON. The 2D drawing comes back, including the scan.

Clear lines deletes the sketch and leaves the handle. **Restore** returns to the original sketch for this session. Neither one is Undo.

10. Chat

Open **Chat** and type the change the way you would say it at the bench. Examples: “put a dagger on the Agent grip”, “remove the guards”, “lengthen the blade half an inch”, “narrow the belly”. Send adds your line and the studio’s reply to the same conversation, so you can keep refining. **+ New chat** starts over. The chat does not replace the drawing tools. If a result is wrong, Undo is still one click on the bar.

A dagger on a grip means the heel through the ricasso stays the grip’s own tang, and the blade forward of the ricasso is the dagger, blended at the ricasso. The grip scales stay the grip you had. Guards on a dagger blank are quillons. “Remove the guards” clips those off. It does not cut the glass breaker off the Agent tang. Everything behind the ricasso stays.

11. A short first session

1. Handle → the grip you are designing for.
2. File → name the blade. Optionally pick a preset.

3. Line → close an outline from the tang, around the tip, and back. Hold Alt to drag a point. Use Circle where the outline is a true radius, and type the radius on the bar.
4. Bevel → two clicks along the edge. Leave the defaults unless the work order says otherwise.
5. Swedge → two clicks on the spine if the tip needs one. Set which end is tall.
6. Chamfer or Edge fillet → two clicks on a corner that should be broken flat or rounded. Set the numbers in the feature panel.
7. Build → wait. Orbit the 3D view. Drag the lamp if the grind is hard to see.
8. File → Save, then Export if someone else needs the design.

12. When something is wrong

WHAT YOU SEE	WHAT TO DO
Build stays dim and shows a number of open loops.	The outline is not closed, or an extra chain is still open. Join the free ends. One closed loop is required.
A choil or ramp sits outside the steel.	Those features cut material away. Place them on the outline, not out in space. Snap on helps.
The scan is upside down.	Mirror, then match the two holes again in the same order.
The scan is close but the junction is off.	Nudge. Rotate, shift, and scale are separate. Undo nudge if you overshoot.
Build says it is still working.	Wait. The status counts seconds. Do not click Build again unless you intend to start over.
STEP or DXF downloads an older blade.	Build again after the last change, then download.
Chrome will not open the page.	The command window must still be running `npm run dev`. The address is `http://127.0.0.1:8080`
A click on an endpoint starts a line when you meant to drag it.	Hold Alt (Option on a Mac) when you click the point, and keep holding it while you drag. A plain click always draws.
The Radius field says the radius is shorter than half the line.	Those two ends are too far apart for that radius. Type a larger radius, or move an end closer.
Tangent says the corner is too square.	One curve cannot leave square to its own ends. Use Fillet on that corner.
The tang end needs to move.	Hold Alt and drag the gold dot at that end.
Trim says the line does not cross another line.	Trim only removes a piece that meets another line. Extend an open end, or delete the line.
Fillet says the radius is longer than the line, or the corner is too flat.	Type a smaller radius in the Fillet field, then click the corner again. If more than two lines meet there, select the two you want first.
An edge fillet does not show on the solid.	It does not change the outline. Click Build after you place it. The lilac mark on the drawing is only the location.
Build says Build123d is not installed.	The STEP kernel is missing. Follow Appendix A, then click Build again.
A new package was just unzipped over the old folder.	Run `npm install` once, then `npm run dev`. Leave the `.cad` folder in place if it is already there.

Appendix A. Install the STEP modeler on Windows or Mac

Build writes a real STEP solid. The modeler is Build123d, which sits on OpenCASCADE. It is the same kernel the shop Linux machine uses. It is not in the zip because the download is large and it has to match the computer it runs on. You install it once, inside the studio folder, in a folder named `.cad`. The studio looks there automatically. On Mac that file is `.cad/bin/python`. On Windows it is `.cad\Scripts\python.exe`.

Use Python 3.12 from python.org. Build123d's geometry library publishes installers for Python 3.11, 3.12, 3.13, and 3.14, on Windows 64-bit and on both Apple Silicon and Intel Macs. Python 3.12 is the one these steps use. Do not use the Python from the Microsoft Store, and do not use the old Python that comes with macOS.

You need an internet connection the first time. The download is a few hundred megabytes. Leave about 2 GB free. The first Build after installing can take a couple of minutes. Later builds are the same speed as on the shop machine.

A.1 Windows

1. Download the Python 3.12 Windows installer (64-bit) from python.org/downloads/windows. Run it. On the first screen, check **Add python.exe to PATH**, then choose Install Now.
2. Unzip the studio if you have not already, and open a command window in that folder. Click the folder's address bar, type `cmd`, and press Enter.
3. Confirm Python answers, then create the modeler beside the studio. Paste this whole block:

```
py -3.12 -m venv .cad
.cad\Scripts\python.exe -m pip install --upgrade pip
.cad\Scripts\python.exe -m pip install build123d pyyaml
```

4. Wait until pip says it successfully installed build123d. A red error about a missing Microsoft C++ compiler means pip tried to compile instead of downloading a ready-made wheel. That usually means the Python is not 3.12 64-bit. Remove the `.cad` folder and repeat from step 1 with the 64-bit 3.12 installer.
5. Check the install:

```
.cad\Scripts\python.exe -c "import build123d, yaml; print('ready')"
```

It should print `ready`. Then start the studio with `npm run dev` if it is not already running, open Chrome at `http://127.0.0.1:8080`, and click Build.

> Windows hides folders that start with a dot, so you may not see `.cad` in File Explorer. That is fine. Do not delete it when you update the studio. If you replace the whole folder with a new zip, run the three commands above again.

A.2 Mac

1. Download the Python 3.12 macOS installer from python.org/downloads/macos. Use the universal installer. It covers Apple Silicon and Intel. Open the pkg and install it. If macOS blocks it, open System Settings → Privacy & Security and allow the installer.

2. Open Terminal. Type `cd ` (with the space) and drag the studio folder onto the window. Press Enter.

3. Create the modeler and install it:

```
python3.12 -m venv .cad
.cad/bin/python -m pip install --upgrade pip
.cad/bin/python -m pip install build123d pyyaml
```

4. Check it:

```
.cad/bin/python -c "import build123d, yaml; print('ready')"
```

It should print `ready`. Then `npm run dev`, Chrome at `http://127.0.0.1:8080`, and Build.

If Terminal says `python3.12: command not found`, the installer did not finish or Terminal was opened before the install. Close Terminal, open it again, and retry. The python.org install also places a copy at `/Library/Frameworks/Python.framework/Versions/3.12/bin/python3`. You can use that full path in place of `python3.12` on the first command only.

A.3 If the install fails

WHAT YOU SEE	WHAT TO DO
`py` or `python3.12` is not recognized.	Python 3.12 is not installed, or the command window was opened before the install. Install it from python.org, close the window, and start again in the studio folder.
pip talks about building a wheel, or asks for a C++ compiler.	The Python is the wrong version or not 64-bit. Delete the `.cad` folder and repeat with Python 3.12 64-bit.
A download stops or times out.	The OpenCASCADE wheel is large. Run the last pip command again. It resumes.
The check does not print `ready`.	Read the last lines of the error. If it cannot import build123d, the pip install did not finish. Run it again.
Build still says Build123d is not installed.	The `.cad` folder is not inside the studio folder, or the studio was started from a different folder. Stop it with Ctrl+C, `cd` to the studio folder, and run `npm run dev` again.

You do not install a separate Linux program, WSL, or a virtual machine. The Windows and Mac packages of this same modeler are what Build calls.